

Machine Design (EMCH327 Section 001, Spring 2022)

Mid-Term Exam 1, 8:30 AM, Friday, Feb. 11th, 2022

Name (First/Last): _____ Score: _____/100'

1. Single choice questions. There is one and only one correct answer for each question. (10 × 3' = 30' in total)

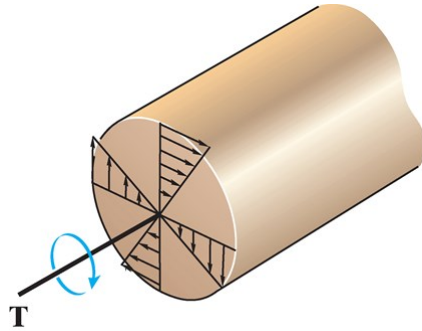
Q1: What is the peak of the stress-strain curve called?

- A. Elasticity of modulus
- B. Ultimate strength

Answer: B

Explanation: On page 28 in the textbook, it says “*The peak of the stress-strain curve is considered the ultimate tensile strength, sometimes called the ultimate strength or simply the tensile strength*”.

Q2: What kind of stress is induced in the cross section by a torque?



- A. Shear stress
- B. Normal stress

Answer: A

Explanation: According to the torsion formula, $\tau = \frac{T\rho}{J}$, shear stress is induced in the cross section by a torque. See Figure 3-9 on page 98 in the textbook.

Q3: What relationship applies to a composite under mechanical deformation, assuming there is no slip at the fiber-matrix interfaces?

- A. The fiber and matrix have same **stress**, i.e., $\sigma_f = \sigma_m$
- B. The fiber and matrix have same **strain**, i.e., $\epsilon_f = \epsilon_m$

Answer: B

Explanation: The fiber and matrix have same strain. The relationship among stresses and strains for a composite and its fiber and matrix materials can be seen in Figure 2-27 on page 72 in the textbook.

Q4: What is the ability of a material to absorb applied energy without failure?

- A. Toughness
- B. Ultimate strength

Answer: A

Explanation: On page 35 in the textbook, it says “*Toughness is the ability of a material to absorb applied energy without failure. Parts subjected to suddenly applied loads, shock, or impact need a high level of toughness.*”

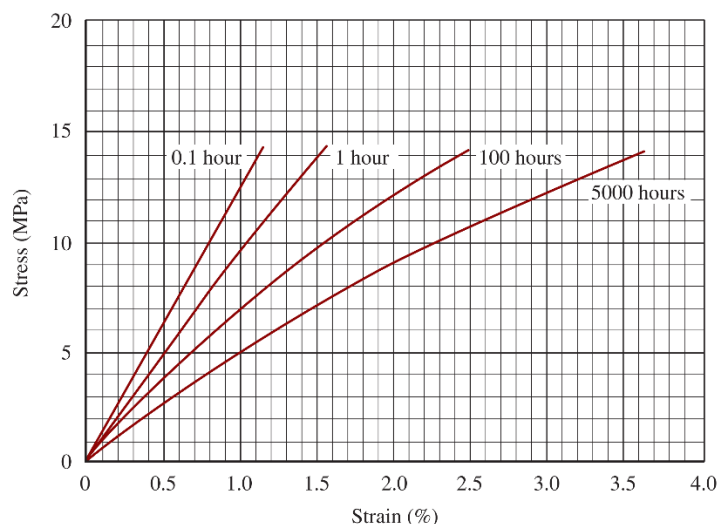
Q5: What is the definition of specific stiffness?

- A. The ratio of the tensile strength of a material to its specific weight
- B. The ratio of the modulus of elasticity of a material to its specific weight

Answer: B

Explanation: On page 67 in the textbook, it says “*specific modulus is the ratio of the modulus of elasticity of a material to its specific weight.*” Because the modulus of elasticity is a measure of the stiffness of a material, the specific modulus is sometimes called specific stiffness.

Q6: What is the corresponding phenomenon for the stress-strain curves as a function of time?



- A. Creep

B. Strain hardening

Answer: A

Explanation: When materials are subjected to high loads continuously, they may experience progressive elongation over time. This phenomenon, called creep. Creep can occur for many plastics even at or near room temperature. Figure 2-9 on page 37 in the textbook shows one way that creep data are displayed for plastic materials.

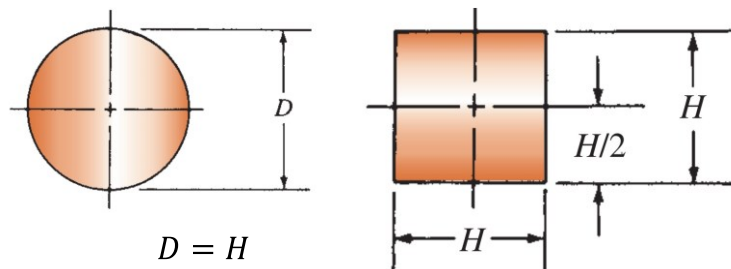
Q7: In which bending test, the highest stressed part of the beam experiences no shearing force?

- A. 3-point bending
- B. 4-point bending

Answer: B

Explanation: On page 31 in the textbook, the bending moment diagram for 4-point bending [Figure 2-5(b)] shows that, the middle region of the beam has the highest bending moment and therefore the highest normal stress but there is no corresponding shearing force.

Q8: If the diameter of a circular cross-section is equal to the side length of a square cross-section, i.e., $D = H$, which cross-section has higher section modulus (S)?

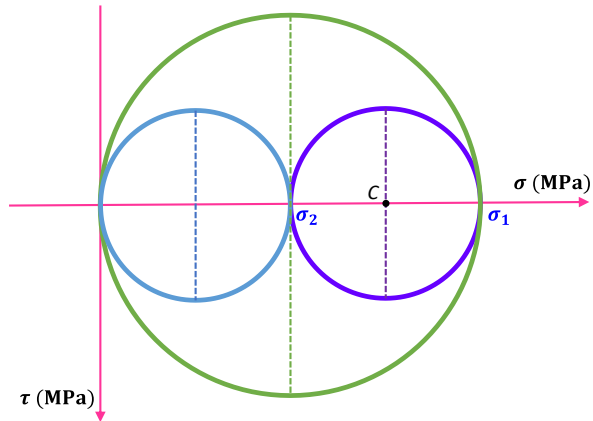


- A. Circular cross-section
- B. Square cross-section

Answer: B

Explanation: According to Appendix 1 in the textbook, the section modulus for a circular and square cross-section is $S_{\text{circular}} = \pi D^3/32$ and $S_{\text{square}} = H^3/6$, respectively. When $D = H$, we have $S_{\text{square}} > S_{\text{circular}}$.

Q9: For a plane stress state, if the three Mohr's circles are drawn in the figure below, then, what is the absolute maximum shear stress ($\tau_{\text{max}}^{\text{abs}}$)?



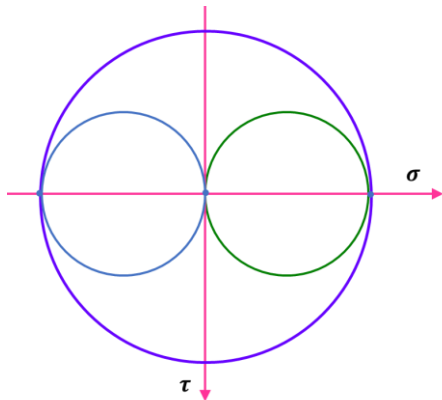
A. $\sigma_1/2$

B. $\frac{\sigma_1 - \sigma_2}{2}$

Answer: A

Explanation: The absolute maximum shear stress ($\tau_{\max}^{\text{abs}}$) is always read from the largest Mohr's circle. For the two plane stresses having same positive sign, the $\tau_{\max}^{\text{abs}}$ is then equal to half of σ_1 .

Q10: What kind of loading induces the plane stress that is represented by the following Mohr's circle (the two inner circles are identical)?



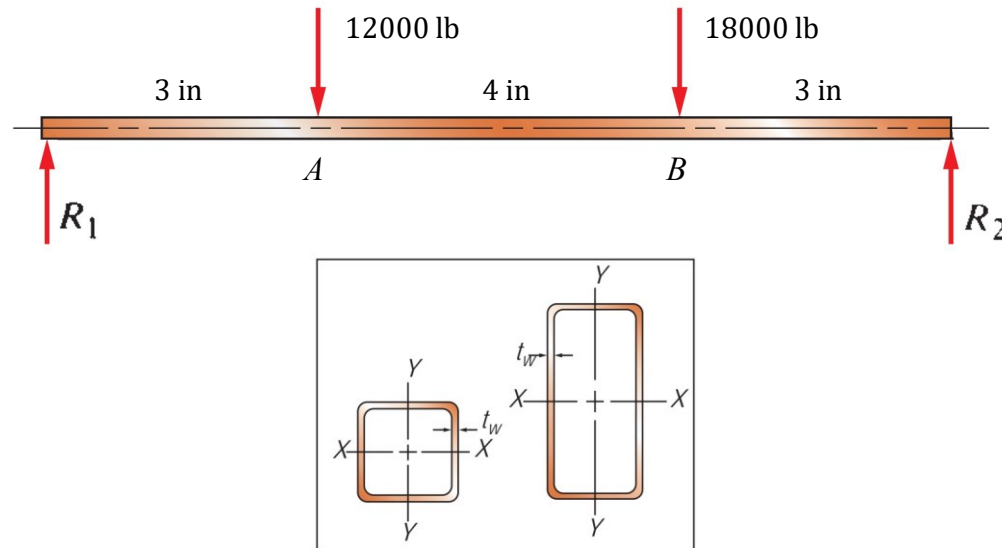
A. Uniaxial tension

B. Pure shear

Answer: B

Explanation: Figure 4-28 on page 161 in the textbook shows the Mohr's circle for pure shear stress state.

2. From Table 15-14 in Appendix 15 in the textbook, select an appropriate cross-section (only need to specify the reference number in the table) for a hollow tubing to carry the load shown below with a maximum normal stress of 2730 psi. The cross-section of the hollow tubing is shown at the bottom and the bending is along the X-X axis. (30' in total)



Solutions:

Step 1: Determine the support reactions (R_1 and R_2) for the loadings by using Free-Body-Diagrams:

$$\sum F_y = 0 \quad \rightarrow \quad R_1 + R_2 = 12,000 + 18,000 = 30,000 \text{ lb} \quad (1)$$

$$\sum M_{R_1} = 0 \quad \rightarrow \quad R_2 \times 10 = 12,000 \times 3 + 18,000 \times 7 = 162,000 \text{ lb} \cdot \text{in} \quad (2)$$

Solving Eqs. (1) and (2) we get:

$$R_1 = 13,800 \text{ lb}$$

$$R_2 = 16,200 \text{ lb}$$

Step 2: plot the shear force and bending moment diagrams using graphical method.

From bending moment diagram, we find that the maximum bending moment happens at position B with magnitude of $48,600 \text{ lb} \cdot \text{in}$.

Step 3: To meet design stress $\sigma_{\max} = 2730 \text{ psi}$, the required section modulus should be at least:

$$S \geq \frac{M_{\max}}{\sigma_{\max}} = \frac{48600 \text{ lb} \cdot \text{in}}{2730 \text{ psi}} = 17.80 \text{ in}^3$$

Step 4: Reading Appendix 15 – 14 we can select

Ref. c with $S = 17.9 \text{ in}^3$.

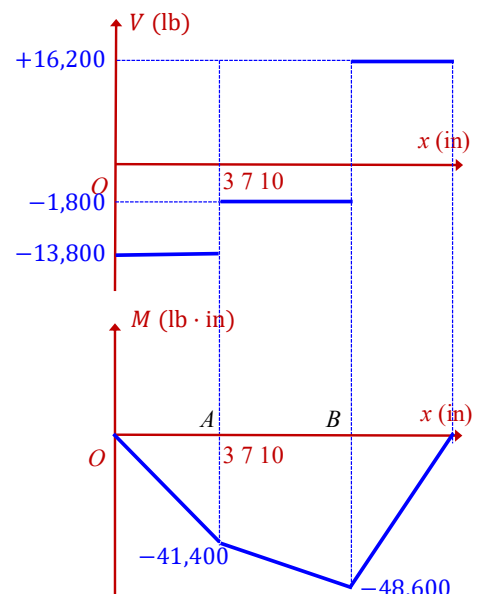
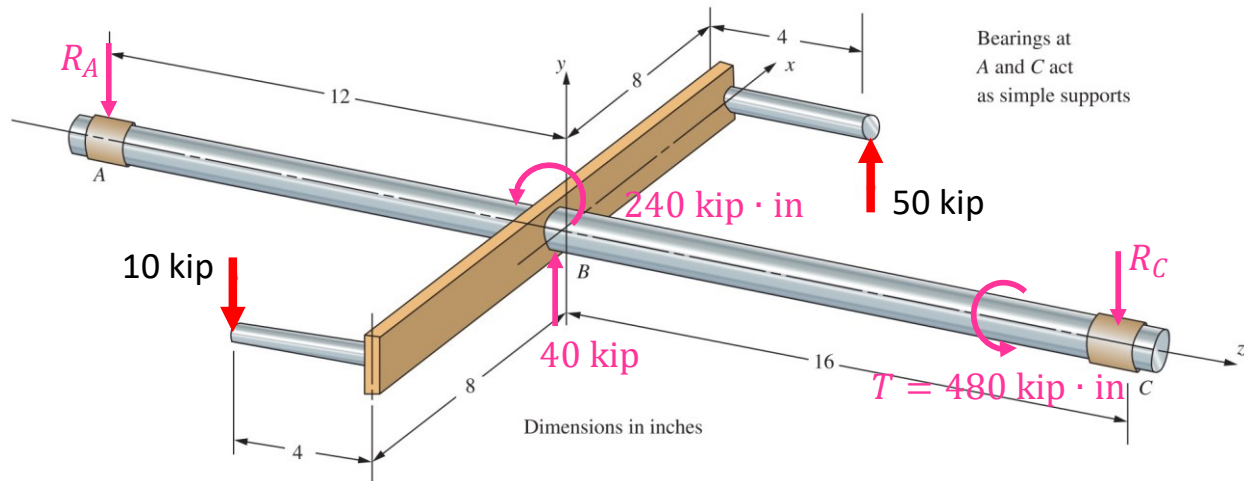


TABLE 15-14 Hollow Tubing: Steel, Square and Rectangular, Standard Structural HSS Shapes: 2 in to 8 in Depth

Ref.	Shape			Design wall thickness t_w (in)	² Weight per foot (lb/ft)	Area A (in ²)	¹ Section properties							
							Axis X-X		Axis Y-Y			Torsional constants		
	(in)	(in)	(in)				I_x (in ⁴)	S_x (in ³)	r_x (in)	I_y (in ⁴)	S_y (in ³)	r_y (in)	J (in ⁴)	C (in ³)
a	HSS 8×8× $\frac{1}{2}$			0.465	48.7	13.5	125	31.2	3.04	125	31.2	3.04	204	52.4
b	HSS 8×8× $\frac{1}{4}$			0.233	25.8	7.10	70.7	17.7	3.15	70.7	17.7	3.15	111	28.1
c	HSS 8×4× $\frac{1}{2}$			0.465	35.1	9.74	71.8	17.9	2.71	23.6	11.8	1.56	61.1	24.4
d	HSS 8×4× $\frac{1}{4}$			0.233	19.0	5.24	42.5	10.6	2.85	14.4	7.21	1.66	35.3	13.6
e	HSS 8×2× $\frac{1}{4}$			0.233	15.6	4.30	28.5	7.12	2.57	2.94	2.94	0.827	9.36	6.35
f	HSS 6×6× $\frac{1}{2}$			0.465	35.1	9.74	48.3	16.1	2.23	48.3	16.1	2.23	81.1	28.1
g	HSS 6×6× $\frac{1}{4}$			0.233	19.0	5.24	28.6	9.54	2.34	28.6	9.54	2.34	45.6	15.4
h	HSS 6×4× $\frac{1}{2}$			0.233	15.6	4.30	20.9	6.96	2.20	11.1	5.56	1.61	23.6	10.1
i	HSS 6×2× $\frac{1}{4}$			0.233	12.2	3.37	13.1	4.37	1.97	2.21	2.21	0.810	6.55	4.70
j	HSS 4×4× $\frac{1}{2}$			0.465	21.5	6.02	11.9	5.97	1.41	11.9	5.97	1.41	21.0	11.2
k	HSS 4×4× $\frac{1}{4}$			0.233	12.2	3.37	7.80	3.90	1.52	7.80	3.90	1.52	12.8	6.56
l	HSS 4×2× $\frac{1}{4}$			0.233	8.78	2.44	4.49	2.25	1.36	1.48	1.48	0.779	3.82	3.05
m	HSS 3×3× $\frac{1}{4}$			0.233	8.78	2.44	3.02	2.01	1.11	3.02	2.01	1.11	5.08	3.52
n	HSS 3×2× $\frac{1}{4}$			0.233	7.08	1.97	2.13	1.42	1.04	1.11	1.11	0.751	2.52	2.23
o	HSS 2×2× $\frac{1}{4}$			0.233	5.38	1.51	0.747	0.747	0.704	0.747	0.747	0.704	1.31	1.41

3. The diameter of the main shaft portion, labeled A , B , and C is 4 in. Determine the maximum tensile stress and absolute maximum shear stress for a point on the bottom of the shaft just **left** to the point B (in the direction to point A). Include any unbalanced torque on the shaft that tends to rotate it about the z -axis. Draw the complete shearing force and bending moment diagrams for loading in the y - z plane. (40' in total)



Solutions:

Step 1: Solve reactions by Free-Body-Diagram analysis:

$$R_A + R_C = 40 \text{ kip} \quad (1)$$

$$R_C \times (16 + 12) = 50 \times 4 + 10 \times 4 + 40 \times 12 \quad (2)$$

Solving Eqs. (1) and (2) we get the reaction forces:

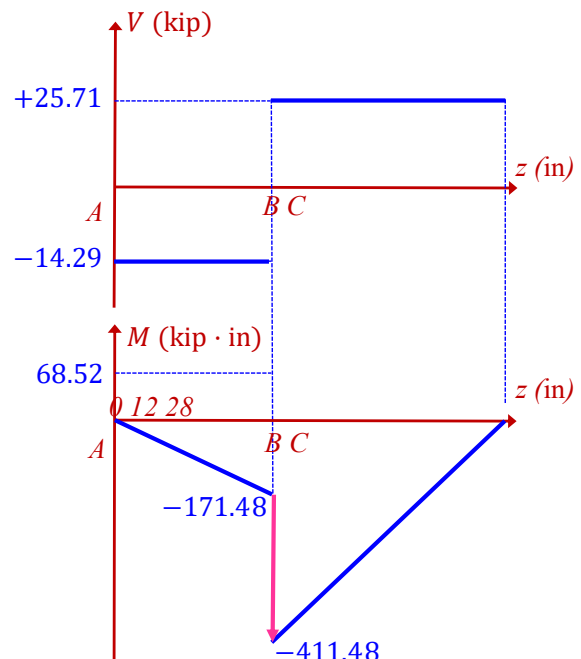
$$R_A = 14.29 \text{ kip}, \quad R_C = 25.71 \text{ kip}$$

Step 2: Plot shear force and bending moment diagrams with graphical method. With moment diagram, we find the maximum bending moment happens at point B with magnitude

$$M_{\max} = 411.48 \text{ kip} \cdot \text{in}$$

Step 3: for a point on the bottom of the shaft just left to the point B , the stress state is plane stress. The tensile stress due to bending moment is

$$\sigma = \frac{M_B}{S} = \frac{M_B}{\pi D^3/32} = \frac{411.48}{\pi 4^3/32} = 65.5 \text{ ksi}$$



The shear stress due to torsion at the same surface point is

$$\tau = \frac{T}{Z_p} = \frac{T}{\pi D^3/16} = \frac{480}{\pi 4^3/16} = 38.2 \text{ ksi}$$

Step 4: Given $\sigma_x = 65.5 \text{ ksi}$, $\sigma_y = 0$, $\tau_{xy} = 38.2 \text{ ksi}$, the center of the circle C is located on the σ -axis, at the point $\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_y}{2} = 32.75 \text{ ksi}$

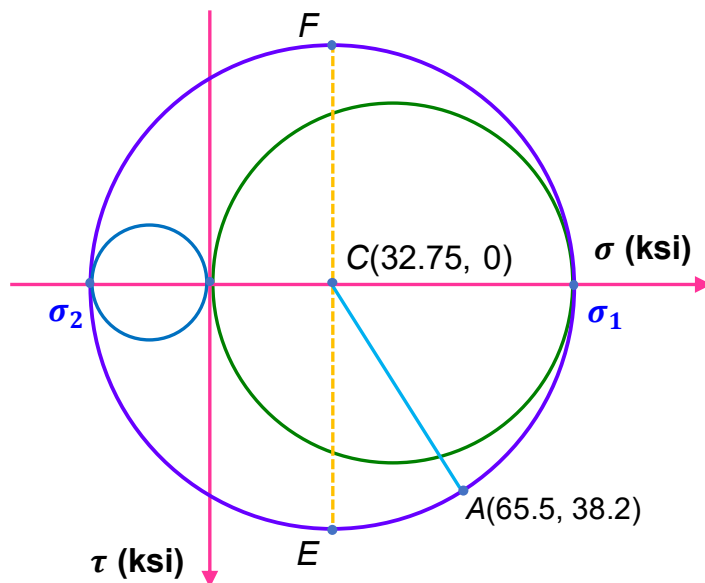
The point $C(32.75, 0)$ and the reference point $A(65.5, 38.2)$ are labeled. Then, the radius of the circle is $R = \sqrt{(32.75 - 65.5)^2 + 38.2^2} = 50.3 \text{ ksi}$, and the circle is constructed.

Using the Mohr's circle, the maximum tensile stress is:

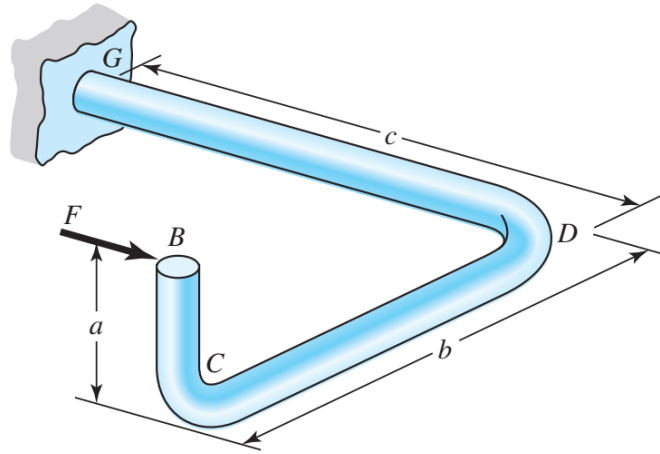
$$\sigma_1 = 32.75 + 50.3 = 83.05 \text{ ksi}$$

The absolute maximum shear stress is:

$$\tau_{\text{max}}^{\text{abs}} = R = 50.3 \text{ ksi}$$



*****Bonus Problem***** Determine the total equivalent bending moment at any cross-section in the segment DG , when a concentrated force $F = 10 \text{ kN}$ is applied to the end B . Take the size $a = 0.5 \text{ m}$, $b = 2 \text{ m}$, and $c = 3 \text{ m}$. The diameter of the wire is 0.1 m . (**10'** in total)



Solutions:

Use the section method to cut through DG and the resultant bending moments and forces can be sketched in the schematic figure.

So, there are two bending moments induced by the concentrated force.

$$M_{DG1} = F \times a = 10 \text{ kN} \times 0.5 \text{ m} = 5 \text{ kN} \cdot \text{m}$$

$$M_{DG2} = F \times b = 10 \text{ kN} \times 2 \text{ m} = 20 \text{ kN} \cdot \text{m}$$

The total resultant bending moment is

$$M_{total} = \sqrt{(M_{DG1})^2 + (M_{DG2})^2}$$

$$= \sqrt{5^2 + 20^2} = \boxed{20.62 \text{ kN} \cdot \text{m}}$$

